Food Loss and Waste Analysis

Where Reported Loss Concentrates: Pareto Analysis

Project Charter

Project goal: To analyse supply chain stages and commodities with the highest recorded food loss in the database (2000 to 2024), and show where the loss is concentrated along the food supply stages.

Project Overview: This project analyses the FAO Food Loss and Waste Database (2000 to 2024) to show where reported food loss concentrates along the supply chain. It uses 21,777 records from 115 countries across 19 supply-chain stages, and the findings are presented in an interactive Tableau Public dashboard. Pareto analysis of high-loss records (reported loss of 20% or more) identifies the food supply chain stages.

Project Scope:

Item	Scope
Data Source	FAO Food Loss and Waste Database, Technical Platform on the Measurement and Reduction of Food Loss and Waste. https://www.fao.org/platform-food-loss-waste/flw-data/en/
Records	21,777 records after exclusions
Measure	Median of reported loss_percentage
Analysis	Charts, Pareto of high-loss records (20% or more)
Deliverables	Tableau Public dashboard, report, README

Data:

- **Source:** FAO Food Loss and Waste Database, Technical Platform on the Measurement and Reduction of Food Loss and Waste. https://www.fao.org/platform-food-loss-waste/flw-data/en/
- **Date accessed:** 23 October 2026.

Data Cleaning log

Step	Rows Left
Raw Download	31,455
Remove exact duplicates on all 18 columns	31,275
Remove rows with a blank supply chain stage	27,884

- Kept 9 of 18 columns and dropped 9 (region, loss_quantity, activity, treatment, cause_of_loss, sample_size, reference, url, notes).
- cpc_code lost its leading zeros on saving; it is not used in the analysis.
- loss_percentage_original mixes two scales; it is not used in the analysis.
- Country names are used as published in the dataset.

The dashboard excludes them with these Tableau filters:

- **Food Supply Stage:** exclude "Whole supply chain"
- **Country:** exclude World and the 13 regional aggregate names (Africa, Asia, Europe, Central Asia, Northern Africa, Northern America, South-Eastern Asia, Southern Asia, Sub-Saharan Africa, Western Africa, Western Asia, Latin America and the Caribbean, Australia and New Zealand).

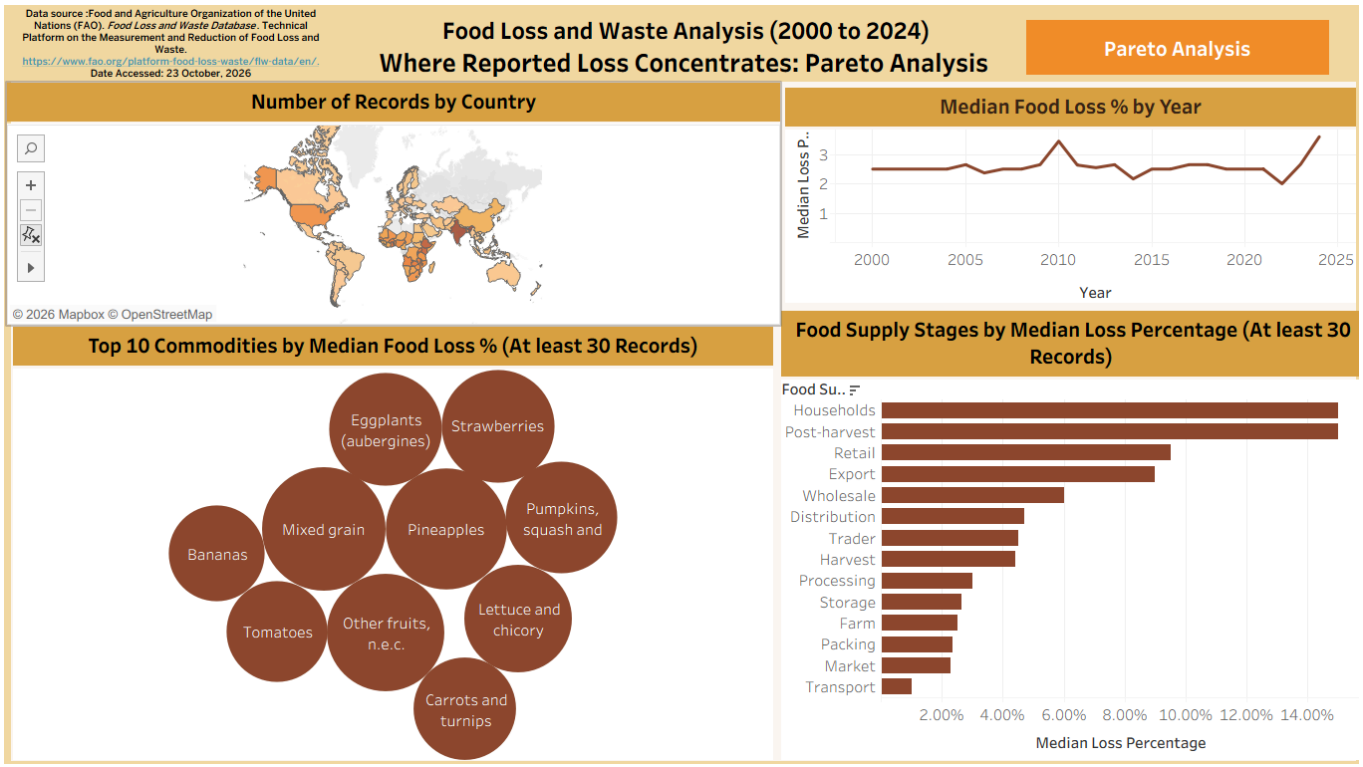
Filter	Records
Cleaned File	27,884
After excluding Whole supply chain, World, and the 13 regional aggregate names.	21,777

Methodology

1. The project was to analyse the supply chain stages and commodities with the highest reported food loss.
2. Data collection: Data were obtained from the FAO Food Loss and Waste Database (31,455 records) (accessed 23 October 2026).
3. Data cleaning: Exact duplicates and records with no supply chain stage were removed, leaving 27,884 rows.
4. Analysis: The records for the whole supply chain and the World and regional aggregates were all excluded to avoid errors. The final set consisted of 21,777 records from 115 countries across 19 stages.

5. Loss % was presented as the median by stage, commodity, and year; record counts were shown by country. The median was used because the values are skewed.
6. Visualisation: Interactive Tableau dashboard.
7. Pareto analysis: the records with a loss of 20% or more were tallied by stage to see where the high loss records were concentrated.
8. Validation: all the figures have been checked against the raw data, and the limitations of the project were noted.

Dashboard- Food Loss and Waste Analysis

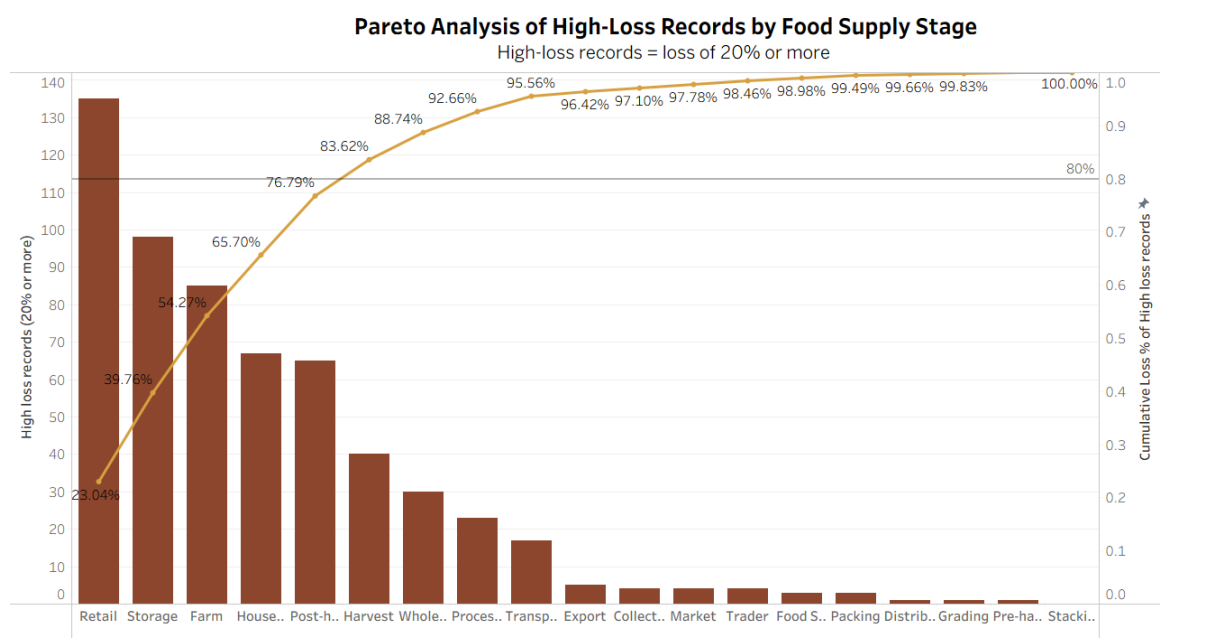


Key Insights

- The median reported loss is 2.5% across 21,777 records.
- Post-harvest and household stages stand out. Both reach a 15.00% median, followed by Retail at 9.50%. Transport records the lowest loss at 1.00%.
- Among commodities with at least 30 records, Mixed grain (12.00%), Pineapples (11.52%), and Other fruits, n.e.c. (10.85%) rank highest.
- The number of records by country is uneven. India (1,233 records), Ethiopia (992), and Kenya (909) dominate, which limits country-level comparison.

Pareto Analysis

A record is classified as high-loss when its reported loss is 20% or more. This is a project-defined threshold, used only for the Pareto analysis.



- High-loss records are concentrated. Out of 586 high-loss records, six stages hold 83.62%: Retail, Storage, Farm, Households, Post-harvest, and Harvest.
- **Retail is the largest contributor.** It accounts for 135 records (23.0%).
- **The 80% line is crossed at the sixth stage.** The first five stages cover 76.79%, and adding Harvest makes the total 83.62%.

Limitations

- Medians are not weighted by production volume, so they show reported loss.
- Some countries, commodities, and stages have few records, so their medians are less stable, and years 2022 to 2024 also have fewer records.
- Loss and waste are reported in a single field (loss_percentage).
- Whole-supply-chain, World, and regional aggregate rows are excluded to avoid double counting.
- A record is classified as high-loss when its reported loss is 20% or more. This is a project-defined threshold, used only for pareto analysis.

Sources

- FAO. Food Loss and Waste Database. Technical Platform on the Measurement and Reduction of Food Loss and Waste. fao.org/platform-food-loss-waste/flw-data.

(Accessed on: 23 October 2026)

Dashboard Link ;

https://public.tableau.com/views/FoodLossandWasteAnalysis/FoodLossandWasteAnalysis

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